CBSE AI Practical 2026-27

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Problem

Diagnostic Gap: The inability of non-specialist health workers to accurately distinguish between complex respiratory sounds leads to high misdiagnosis rates.

Inaccessible Technology: Advanced digital stethoscopes cost upwards of ₹35,000, making AI diagnostic tools unaffordable for local clinics and rural PHCs.

The "Trial-and-Error" Risk: Local doctors often prescribe generic medication for serious symptoms, wasting critical time while underlying infections like Pneumonia worsen.

Critical Referral Delays: lack of early, objective detection for high-risk conditions results in late-stage hospitalizations and increased mortality in respiratory cases.

The Opportunity

Real-time AI classification allows local doctors to identify high-risk pathologies instantly, ensuring life-saving treatment begins days earlier.

Using low-cost, off-the-shelf hardware and open-source AI, this solution can be rapidly deployed across global regions with limited medical resources.

Converting analog breath sounds into digital data enables the creation of a longitudinal health record for better patient monitoring and follow-up.

By reducing the cost of digital stethoscopes by 80%, we can equip every ASHA worker and rural clinic with specialist-level screening tools.

Our Idea

1. VITALIS uses an embedded neural network to instantly translate raw lung sounds into clinical labels like "Normal," "Wheeze," or "Crackle."

2. By utilizing a high-fidelity I2S MEMS microphone, the device captures crystal-clear respiratory data without the electrical noise of traditional analog sensors

3. All AI inference is performed locally on the XIAO C6 microcontroller, ensuring diagnostic results are available in under 6ms without needing an internet connection.

4. The system provides a standardized "second opinion" to local healthcare workers, reducing human error and ensuring high-risk patients are referred to hospitals immediately.

Benefits

Elimination of Diagnostic Subjectivity

Replaces inconsistent human hearing with a standardized AI model, ensuring every patient receives an objective and accurate lung assessment.

Prevention of Critical Complications

Enables early detection of high-risk pathologies, allowing for timely hospital referrals and reducing the risk of minor infections turning into emergencies.

Cost-Effective Healthcare Infrastructure

Utilizes affordable, off-the-shelf hardware to provide professional-grade digital stethoscopy at a fraction of the cost of current market alternatives.

Enhanced Rural Medical Reach

Empowers frontline health workers and ASHA volunteers to perform complex diagnostic screenings in remote areas without specialist presence.

5W1H

3. 5W1H ANALYSIS

Question	Specification
Who?	ASHA Workers, NGO volunteers, and PHC medical officers across rural India.
What?	An Edge-AI hardware attachment that digitizes and analyzes lung acoustics in real-time.
Where?	Focused on Primary Health Centers (PHCs) and Community Health Centers (CHCs) in underserved districts.
When?	During the first point of contact in medical camps or rural check-ups.
Why?	To reduce infant mortality rates (IMR) caused by late-detected Pneumonia and manage high COPD prevalence.
How?	Through Digital Signal Processing (DSP) and int8 quantized Neural Networks running on the ESP32-C6.

Thank You